Algorithmic Accountability and Operational Efficacy: A Technical Audit of k-Nearest Neighbors in Public Sector Applications

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***Abstract*— This report presents a comprehensive technical and operational analysis of the k-Nearest Neighbors (kNN) algorithm, evaluating its utility in high-stakes government decision-making. Building upon a baseline predictive modeling exercise using the Fisher Iris dataset, this study expands the scope to real-world public sector applications documented in literature from 2021 to 2026. Specifically, we investigate the deployment of kNN in healthcare fraud detection (USA), urban land-use zoning (South Africa), and disaster response prediction (Indonesia). The analysis reveals that while kNN offers superior interpretability ("glass-box" transparency) compared to deep learning models, its effectiveness is highly sensitive to environmental factors such as data dimensionality and class imbalance. A deep-read technical audit of a flood risk prediction system highlights the critical role of preprocessing and hyperparameter tuning (k-value selection) in achieving operational reliability. Finally, we synthesize these findings to address the ethical implications of algorithmic decision-making, arguing that kNN's transparency makes it a vital tool for due process, despite its performance limitations in high-volume environments.**

Keywords*— *k-nearest neighbors (knn), predictive modeling, fisher iris dataset, public sector applications, healthcare fraud detection, urban land-use zoning, disaster response prediction, interpretability, data dimensionality, class imbalance, preprocessing, hyperparameter tuning, flood risk prediction, algorithmic decision-making, ethical implications

I. Introduction

The k-Nearest Neighbors (kNN) algorithm remains a cornerstone of supervised machine learning due to its intuitive "lazy learning" architecture, where generalization is deferred until the query phase.[1] While recent advancements in Artificial Intelligence (AI) have shifted toward opaque Deep Neural Networks (DNNs), kNN retains critical relevance in the public sector where "right to explanation" and algorithmic transparency are legal mandates.[2]

This report serves two purposes: first, to contextualize the findings of a baseline Iris classification experiment within the broader state-of-the-art; and second, to answer specific operational questions regarding the deployment of kNN in government agencies. By analyzing literature published between 2021 and 2026, we validate the algorithm's performance across diverse domains and scrutinize the technical specifications required for its successful implementation..

II. Real-World Applications & Technical Validation

A. Operational Investigation

We investigated three distinct government operations where kNN and related distance-based algorithms are currently deployed:

1. **Healthcare Fraud Detection (Medicare, USA):**
   - **Agency/Operation:** The Centers for Medicare & Medicaid Services (CMS) and associated research bodies utilize machine learning to combat waste and abuse. Recent studies (2025) focus on detecting specific fraud schemes within Medicare Part B claims [3],[4]
   - **Categorical Predictions**: The primary prediction is binary classification: Fraudulent vs. Legitimate claims. Specific sub-categories predicted include "Upcoding" (billing for more expensive procedures than performed) and "Unbundling" (billing bundled procedures separately to increase reimbursement).[3]
   - **Role of kNN**: kNN is employed to identify "peer groups" of providers. It flags anomalies by calculating the distance between a suspect provider's billing patterns and the average of their **k** nearest peers.[4]
2. **Urban Land-Use Zoning (Cape Town, South Africa):**
   - **Agency/Operation**: Municipal planning and environmental protection agencies in Cape Town utilize satellite imagery to monitor rapid urbanization and informal settlement growth.[5]
   - **Categorical Predictions**: The system performs multi-class classification of satellite pixels into Land Use/Land Cover (LULC) categories: Built-up (residential/commercial), Vegetation, Waterbodies, and Trees.[5]

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- **Role of kNN**: The algorithm maps spectral signatures from satellite imagery (PlanetScope) to these categories, allowing the city to track unauthorized expansion into protected zones.[5]

3. **Disaster Response Prediction (Jakarta, Indonesia):**
   - **Agency/Operation**: The Jakarta Regional Disaster Management Agency (BPBD) utilizes predictive models to manage flood risks in the capital province.[6]
   - **Categorical Predictions**: The model classifies specific administrative regions (RWs) into risk levels: Low, Moderate, or High Risk of flooding based on historical water levels and rainfall intensity.[6]
   - **Role of kNN**: It serves as an early warning system, predicting future flood events by finding historical weather patterns that are "nearest neighbors" to current conditions.[7]

B. Performance & Environmental Factors

The effectiveness of kNN varied significantly across these domains, driven by external environmental factors:

- **Data Quality and Imbalance (Healthcare)**: In the Medicare domain, kNN struggled as a standalone classifier. A 2025 study reported that while ensemble models like Random Forest achieved 98.8% accuracy, kNN achieved only 79.2%.[4] The external factor was Class Imbalance; actual fraud cases are rare (<0.1% of claims), causing kNN to bias predictions toward the majority "Legitimate" class unless rigorous resampling (SMOTE) is applied.[4]
- **Dimensionality and Scale (Disaster Response)**: In Jakarta, kNN performed exceptionally well, with accuracy ranging from 96% to 99%.[6] The success was attributed to the low dimensionality of the data (specific weather parameters) and the cyclical nature of seasonal flooding, which favors distance-based memory models.
- **Computational Constraints (Urban Planning)**: In Cape Town, kNN (96.54% accuracy) actually outperformed Support Vector Machines (SVM) and Random Forest.[5] The environmental factor here was the high resolution of PlanetScope imagery, which created distinct, compact clusters in the feature space that kNN could easily separate without complex training overhead.

III. Technical Specification Audit (Deep-Read)

Primary Source Selected: Ino, A. R., & Taufik, A. (2025). "Flood Risk Prediction Using the K-Nearest Neighbors (KNN) Method with Machine Learning Optimization in Jakarta Province." [6]

A. Distance Metric

Metric Used: The researchers utilized **Euclidean Distance** (often defaulted in Python libraries) and tested **Minkowski Distance** configurations [6],.[7]

Justification: The data consisted of continuous numerical environmental variables (water levels in cm, rainfall intensity). Euclidean distance is appropriate here because it calculates the direct spatial magnitude between the current weather vectors and historical vectors. Unlike text data (where Cosine similarity is preferred), physical measurements of flood depth map well to Euclidean geometry [1],.[6]

B. Preprocessing

- **Method**: The study explicitly implemented Z-Score Normalization and Min-Max Scaling.[6]
- **Logic & Comparison**: The "Why" logic matches the notebook's requirement for feature scaling. Flood data contains features with vastly different scales: "Rainfall" might range from 0-300 mm, while "Flood Frequency" might range from 0-5 events. Without normalization, the "Rainfall" feature would mathematically dominate the distance calculation, rendering the frequency data irrelevant. The authors standardized these to ensure every feature contributed equally to the risk score.[6]

C. k-Value Justification

- **Chosen k:** The study identified **k=3** as the optimal value for the highest accuracy.[6]
- **Method of Selection**: The researchers did not guess; they used **5-Fold Cross-Validation** to test **k** values ranging from 1 to 7.[6] This involves splitting the data 5 times, training on 4 parts and testing on 1, to ensure the chosen **k** is robust and not just lucky for one specific data split. This aligns with the "Bias-Variance Tradeoff" concept, where **k=3** provided a balance between capturing local flood pockets (low bias) and ignoring sensor noise (low variance).[6]

IV. Ethical Decision-Making

The "Black Box" vs. "Glass Box" Dilemma

As a government official, I would feel comfortable relying on kNN for these high-stakes decisions, specifically because of its transparency relative to "Black Box" models like Neural Networks.

- **Algorithmic Transparency:** In the context of the prompt's Confusion Matrix (Task 4c), we saw that kNN made a specific error: misclassifying Versicolor as Virginica. Because kNN is a "Glass Box" [8], we can audit this error. We can query the system: "Show me the 5 neighbors that caused this vote." If the system shows 3 Virginica neighbors that are edge cases, a human expert can validate whether the error was reasonable.[2]
- **Justification:** In high-stakes operations like Healthcare Fraud, denying a provider's payment

based on a Neural Network's opaque "99% probability" is legally risky and ethically fraught. However, denying payment based on a kNN analysis allows the government to say: "We flagged this claim because it is statistically identical to these 5 specific past cases of proven fraud." This adherence to the legal standard of **precedent** makes kNN uniquely ethically defensible in the public sector, provided that the training data itself is audited for historical bias [2],.[8]

VI.. Conclusion

The analysis confirms that kNN is not merely a pedagogical tool but a functional asset in government technology stacks. Its deployment in Jakarta and Cape Town demonstrates its superiority in handling spatial and environmental data where "history repeats itself." While it faces challenges in high-dimensional fraud detection [4], its role as an interpretable baseline is indispensable. Future implementation must rigorously apply the preprocessing (Scaling) and validation (Cross-Validation) techniques identified in the technical audit to ensure reliability.

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